

Thomson jumping ring — Solution

Y23 (2023) — English translation

A1

0.30 pts

Write down an equation relating the current in the ring $I(t)$ and the magnetic field in the core $B_z(t)$.

Let's write Faraday's law of electromagnetic induction:

$$\mathcal{E}_{ind} = -\frac{d\Phi}{dt}$$

For выбранного направления currenta в кольце (против часовой стрелки for/at виде from above) induced EMF equals:

$$\mathcal{E}_{ind} = -\frac{\pi d^2}{4} \dot{B}_z$$

Apply Kirchhoff's second rule to the selected circuit and get:

$$L\dot{I} + IR = -\frac{\pi d^2}{4} \dot{B}_z$$

A2

0.50 pts

Obtain the dependence of the eddy current strength $I(t)$ on time at $B_z(t) = B_m \cos \omega t$.

Let's imagine the magnetic field induction in complex form:

$$B_z(t) = B_m e^{i\omega t}$$

Then:

$$I(i\omega L + R) = -\frac{\pi d^2 B_m}{4} \frac{de^{i\omega t}}{dt} = \frac{\pi d^2 \omega B_m e^{i(\omega t - \pi/2)}}{4}$$

Since $R + i\omega L = \sqrt{R^2 + \omega^2 L^2} \cdot e^{i \arctan(\omega L/R)}$, for $I(t)$ we find:

$$I(t) = I_0 \cos(\omega t - \phi) = \frac{\pi d^2 \omega B_m}{4\sqrt{R^2 + \omega^2 L^2}} \operatorname{Re} \left(e^{i(\omega t - \pi/2 - \arctan(\omega L/R))} \right)$$

From:

Answer: \textbf{Answer}:

$$I(t) = \frac{\pi d^2 \omega B_m}{4\sqrt{R^2 + \omega^2 L^2}} \cos \left(\omega t - \frac{\pi}{2} - \arctan \frac{\omega L}{R} \right)$$

A3

0.30 pts

From Gauss's theorem, find the radial component of the field $B_{0\rho}(z)$. Further, in all points, use the result obtained.

Let us apply the Gauss theorem on the flux of magnetic field induction to a cylindrical surface coaxial with the z axis:

$$\frac{\partial \Phi_{\text{OSH}}}{\partial z} + \frac{\partial \Phi_{\text{OOK}}}{\partial z} = 0$$

Since зависимость $B_z(\rho)$ внутри сердечника можно пренебречь, последнее equation можно записать в следующем виде:

$$\frac{\pi d^2}{4} \frac{\partial B_{0z}(z)}{\partial z} + \pi d \cdot B_{0\rho}(z) = 0$$

From here:

Answer: \textbf{Answer:}

$$B_{0\rho} = -\frac{d}{4} \frac{\partial B_{0z}(z)}{\partial z}$$

A4

0.20 pts

Find the dependence of the axial component of the force $F_z(z, t)$ acting on the ring on the coordinate z and time t .

Let us determine the projection of the Ampere force on the axis z :

$$F_z = -\pi d I B_\rho$$

Then we have:

Answer: \textbf{Answer:}

$$F_z = \frac{\pi^2 d^4 \omega B_{0z}(z)}{16 \sqrt{R^2 + \omega^2 L^2}} \frac{\partial B_{0z}(z)}{\partial z} \cdot \cos(\omega t) \cos(\omega t - \phi), \quad \phi = \arctg \frac{R}{L\omega}$$

A5

0.60 pts

Find $F_z(z)$.

Let's use the formula for the product of cosines:

$$\cos \alpha \cos \beta = \frac{\cos(\alpha - \beta) + \cos(\alpha + \beta)}{2} \Rightarrow \cos(\omega t) \cos(\omega t - \phi) = \frac{\cos(\phi) + \cos(2\omega t - \phi)}{2}$$

Усреднение по времени for/атводит к выражению:

$$\langle \cos(\omega t) \cos(\omega t - \phi) \rangle = \frac{\cos \phi}{2} = -\frac{\sin \left(\arctan \frac{\omega L}{R} \right)}{2} = -\frac{\omega L}{\sqrt{R^2 + \omega^2 L^2}}$$

from where we find for the time-averaged force:

Answer: \textbf{Answer:}

$$F_z(z) = -\frac{\pi^2 d^4 B_{0z}(z)}{32} \frac{\partial B_{0z}(z)}{\partial z} \cdot \frac{\omega^2 L}{R^2 + \omega^2 L^2}$$

B1

0.50 pts

Find an expression for $F_z(z)$ in the geometry under consideration and the potential of this force $U_F(z)$ (assume it is zero at the top of the core).

For the model described in the condition:

$$\frac{\partial B_{0z}(z)}{\partial z} = -\frac{B_0}{H} = \text{const}$$

Hence we get expression for силы $F_z(z)$:

For the potential taking into account $U_F(H) = 0$ we have:

$$U_F(z) = - \int_H^z F_z(z) dz = - \frac{\pi^2 d^4 B_0^2}{32H} \cdot \frac{\omega^2 L}{R^2 + \omega^2 L^2} \int_H^z \left(1 - \frac{z}{H}\right) dz = \frac{\pi^2 d^4 B_0^2}{32} \cdot \frac{\omega^2 L}{R^2 + \omega^2 L^2} \int_0^{1-z/H} t dt$$

from where:

Answer: \textbf{Answer:}

$$F_z(z) = \frac{\pi^2 d^4 B_0^2}{32H} \cdot \frac{\omega^2 L}{R^2 + \omega^2 L^2} \left(1 - \frac{z}{H}\right)$$

B2

0.50 pts

At what height h will the ring float in a state of equilibrium if the field is "turned on" slowly? Express your answer in terms of $m, g, R, L, \omega, d, B_0, H$.

With a slow change in the current strength in the solenoid, the ring will always be located near the equilibrium position, therefore the height of its levitation is determined by the condition of its equilibrium:

$$F_z = mg$$

Substituting expression for $F_z(z)$ and taking into account that в равновесии $z = h$, we find:

$$h = H \left(1 - \frac{32mgH(R^2 + \omega^2 L^2)}{\pi^2 d^4 B_0^2 \omega^2 L}\right)$$

Данный answer for/atменим, if quantity в скобках положительна, иначе $h = 0$, therefore finally:

Answer: \textbf{Answer:}

$$h = \begin{cases} H \left(1 - \frac{32mgH(R^2 + \omega^2 L^2)}{\pi^2 d^4 B_0^2 \omega^2 L}\right) & \text{при } \frac{32mgH(R^2 + \omega^2 L^2)}{\pi^2 d^4 B_0^2 \omega^2} < 1 \\ 0 & \text{при } \frac{32mgH(R^2 + \omega^2 L^2)}{\pi^2 d^4 B_0^2 \omega^2} > 1 \end{cases}$$

B3

0.50 pts

To what height h_m will the ring jump if the field is "turned on" quickly? Express your answer in terms of $m, g, R, L, \omega, d, B_0, H$.

Let us obtain the dependence of the resultant Ampere forces and gravity acting on the ring on its coordinate z :

$$R_z = F_z - mg = \frac{\pi^2 d^4 B_0^2}{32H} \cdot \frac{\omega^2 L}{R^2 + \omega^2 L^2} - mg - \frac{\pi^2 d^4 B_0^2 z}{32H^2} \cdot \frac{\omega^2 L}{R^2 + \omega^2 L^2} = A - kz$$

Это линейная function, положение которой соответствует значению $z_0 = h = A/k$, найденному в предыдущем parte, moreover if $A < 0$ - кольцо так же не подпрыгнет, т.е $h_m = 0$. If в положении равновесия potentialную энергию кольца в поле всех сил for/atнять equals нулю - её зависимость от z имеет следующий вид:

$$U(z) = \frac{k(z - z_0)^2}{2}$$

Кольцо останавливается at the point, в которой его potential energy та же, что and в начальном положении, hence:

$$z_{\text{ост}} = 2z_0 \Rightarrow h_m = 2h = 2H \left(1 - \frac{32mgH(R^2 + \omega^2 L^2)}{\pi^2 d^4 B_0^2 \omega^2 L}\right)$$